# Muhammad Usama Mughal


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## Professional Summary


**Full-Stack Mobile Application Developer** specializing in cross-platform engineering with **Flutter**, native **Android (Kotlin)**, and enterprise backends built on **C# .NET Core & SQL Server**. Proven track record of architecting user-centric mobile applications, designing RESTful APIs, and optimizing database performance. Blends hands-on engineering experience at **Technology Junction (NASTP)** with a strong background in ERP systems, workflow management, and enterprise backend architectures.

- **Core Expertise:** Cross-Platform App Development (Flutter/Dart), C# ASP.NET Core Web APIs, SQL Server, State Management, and Data Caching.
- **Project Highlights:** Designed and integrated the full frontend and C# backend API for **Drovby** (fleet management) and engineered **Accura Vault** (financial analytics tracker).
- **Enterprise & Systems:** Experienced in ERP logic integration, database architecture, server maintenance, and infrastructure operations in high-availability environments.


## TECHNICAL SKILLS

- **Mobile & Cross-Platform:** Flutter (Dart), Kotlin, Java, Jetpack Compose, XML Layouts, Material Design 3, Coroutines, Flow, State Management
- **Backend & Web APIs:** C#, .NET Core, ASP.NET Core Web API, RESTful APIs, Entity Framework Core, Custom Licensing & Auth Logic
- **Architecture & Patterns:** Clean Architecture, MVVM, Repository Pattern, Dependency Injection (Hilt/Koin), Data Caching Architecture
- **Databases & Cloud:** SQL Server, SQLite, Room DB, Firebase (Realtime DB, Auth, Analytics), Google Cloud Console
- **Security & Systems:** MD5 Hashing, Anti-Clock Tamper Logic, Time-Block Validation, ProGuard/R8, ERP Workflows, Server & Linux/Windows Admin

- **UI/UX & Design:** User-Centric Design, Responsive & Adaptive Layouts, Wireframing & Prototyping (Figma), Lottie Animations, Dark Mode
- **Tools & DevOps:** Docker, Git & GitHub, Android Studio, VS Code, Postman, Figma, Adobe Premiere Pro & After Effects

## Professional Experience

### *Software Development Intern | Technology Junction (NASTP)*

*May 2026 – Present | Rawalpindi, Pakistan*

- Engineered robust cross-platform mobile solutions using **Flutter** and **Dart**, focusing on high-performance dynamic UIs and state management.
- Architected and integrated local **C# ASP.NET Core Web APIs** and **SQL Server** database schemas for enterprise operations.
- Implemented client-side data caching strategies (DataCache) and parallel network request handling to optimize app loading times and offline resilience.
- Collaborated on repository setups, EF migration workflows, and backend API endpoints using Git and Docker environments.

### *IT Support Trainee | Pearl Continental Hotel*

*Apr 2025 – Jul 2025 | Rawalpindi, Pakistan*

- Provided technical support for hotel systems, server backups, network troubleshooting, and hardware maintenance.
- Conducted regular inspections of server and exchange rooms to maintain high uptime and operational stability.
- Handled technical setups for corporate meetings, executive guest events, and departmental trainings.

### *Video Editor & Content Researcher | Hashtag Media*

*Dec 2021 – Jan 2025 | Islamabad, Pakistan*

- Produced high-quality visual content using Adobe Premiere Pro and After Effects for digital media projects.
- Led content strategy research to optimize engagement and collaborated with multidisciplinary teams under tight deadlines.

## Featured Projects (Enterprise & Work)

### *Drovby — Vehicle Fleet Management App*

*Technologies: Flutter, Dart, C# ASP.NET Core, SQL Server, Entity Framework Core, REST API*

- Built the frontend UI/UX and integrated backend REST APIs for a full-scale fleet management platform.
- Developed an interactive **Garage Command Center** featuring live maintenance counters, dynamic status pills, and vehicle metrics.
- Handled state-driven network operations, image upload pipelines, and dynamic vehicle data caching.

## Independent Development Projects

### *Al-Muqeet Milk | Business Management Ecosystem (Market Running)*

*Jan 2026 – Present | Rawalpindi, Pakistan*

**Lead Developer | Android Studio Narwhal Feature Drop | 2025.1.2 Patch 2, Kotlin/Java**

- **Automated Ledger Systems:** Engineered a dynamic billing engine within the **Android Studio** environment that generates professional PDF invoices and manages complex customer ledgers automatically.
- **Resilient Backup:** Integrated a hybrid synchronization model using local storage and **Firebase** for cloud-based data redundancy, utilizing **Logcat** and **Firebase Debug View** to ensure data integrity during sync.
- **Proprietary Security:** Leveraged **Android Studio's** advanced debugging and obfuscation tools to develop a robust, offline-capable License Management System using **Time-Block MD5 Hashing**, preventing unauthorized redistribution and data-clearing bypasses.
- **WhatsApp Integration:** Built a direct communication bridge for sending digital bills and notifications to customers via WhatsApp.

### *Aequs Vault | Income/Expense Tracker*

**Lead Developer | Android Studio, Kotlin/Java**

### Technical Core & Logic Implementation

- **Financial Data Aggregation Engine:** Engineered real-time logic to process and synchronize core financial metrics, including **Total Balance**, **Monthly Expenditure**, and **Consolidated Savings** using state-driven updates in Jetpack Compose.
- **Dynamic Progress Tracking:** Developed a mathematical tracking system for "Savings Goals" that calculates real-time percentage completion and visualizes milestones based on dynamic user input and historical data.
- **Expression Chaining Math Engine:** Architected a custom "Quick-Math" terminal supporting complex multi-step equations (e.g., $3+2+2-4$) via intermediate variable management, ensuring mathematical accuracy during long-string operations.
- **Smart Transaction Categorization:** Designed a filtering algorithm to automatically sort high-volume transaction data into **Income** and **Expense** streams, providing users with granular visibility into cash flow.

### Architectural & UI Achievements

- **Advanced UI Design (Clean Architecture):** Leveraged Material 3 and Jetpack Compose to build a high-contrast visual hierarchy, utilizing strategic color coding to differentiate financial statuses for instant data recognition.
- **Currency & Exchange Module:** Developed a dedicated exchange interface utilizing mathematical ratio logic for currency conversion, demonstrating proficiency in handling complex numeric computations and external data integration.
- **Data Reliability:** Ensured high data integrity through local persistence logic (Room Database) and manual cloud-backup routines to protect sensitive financial records against data loss.

## Soft Skills

- **Problem Solving:** Translating complex business rules into performant logic (e.g., custom cryptographic hashing).
- **AI-Assisted Workflow:** Leveraging modern AI tools to accelerate code generation, debug stack traces, and optimize execution speed.
- **Cross-Functional Collaboration:** Communicating effectively across project stakeholders, clients, and technical teams.
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## Education

Intermediate in Computer Science (ICS)
Islamabad Model Postgraduate College of Commerce, H8-4, Islamabad

Matriculation (Science)
BUDS School System (English Medium)

## Certifications

• 3-Month IT Training Certificate – Pearl Continental Hotel, Rawalpindi
• 3-Year Experience Letter – Hashtag Media (Video Editing & Research)

## Personal Interests

- **Advanced Security:** Exploring cryptographic implementations in mobile environments.
- **Automotive Tech:** Following the tech stacks of BMW and Tesla.
- **Fitness:** Active in Football and Gym activities.

## Languages

English – Fluent
Urdu – Native